classmate
Date 22/03/26
Page 1

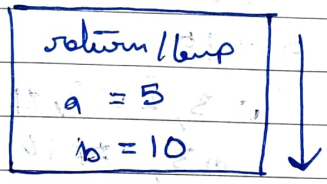
R

RTOS : Assignment -1

(1) C-code

```
int main(void)
{
    int a=5;
    int b=10;
    if(a==0)
        a=a+10;
    b=a;
}
```

Stack Frame :



Data movement

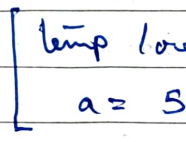
[sp+4](a) → LDR R0 → STR → (sp)(b)

```
// int a = 5
movs  r3, #5
str   r3, [r7, #4]
// int b = 10
movs  r3, #10
str   r3, [r7, #0]
// if(a==0)
ldr   r3, [r7, #4]
cmp   r3, #0
bne   la <main + 0x10>
// a = a+10
ldr   r3, [r7, #4]
adds  r3, #10
str   r3, [r7, #4]
// b = a
ldr   r3, [r7, #4]
str   r3, [r7, #0]
movs  r3, #0
```

② If else cond

```
int main(void)
{
  int a = 5
  if (a==0)
    a = a+10
  else
    a = a+1
  return 0
}
```

Stack frame:



Assembly:

```
// a = 5
movs   r3, #5
str    r3, [r7, #4]
//if (a==0)
ldr    r3 [r7, #4]
cmp    r3, #0
bne.n  18 <main+0x10>
//a = a+10
ldr    r3 [r7, #4]
adds   r3, #10
str    r3, [r7, #4]
b.n    1e <main+0x1e>
// a = a+10
ldr    r3, [r7, #4]
adds   r3, #10
str    r3 [r7, #4]
b.n    1e <main+0x1e>
// a=a+1
ldr    r3, [r7, #4]
adds   r3, #1
str    r3 [r7, #4]
return 0;
movs   r3, #0
```

Data movement

[SP] (a) —LDR→ R0 = 5 ⇒ a = 6

↓

R0 = 6

↓

[SP] (a) = 6

return 0

(3) Recursion

```
int rec_sum(int i)
{
  if (i<1)
    return i;
else
  return i+rec_sum(i-1);
}
int main(void)
{
  int sum = 0;
  sum = rec_sum(10);
  return 0;
}
```

```
// if (i < 1)
    ldr   r3, [r7, #4]
    cmp   r3, #1
    bgt.n 12 <rec_sum+0x10>
// return i;
    ldr   r3, [r7, #4]
    b.n   22 <rec_sum+0x22>
//return i + rec_sum(i-1)
    ldr   r3, [r7, #4]
    subs  r3, #1
    mov   r0, r3
    bl    0 <rec_sum>
    mov   r2, r0
    ldr   r3, [r7, #4]
    add   r3, r2
```

=> int main(void)

```
// int sum = 0
    movs  r3, #0
    str   r3, [r7, #4]
// sum = rec_sum(10)
    movs  r3, #0
    str   r3, [r7, #4]
// sum = rec_sum(10)
    movs  r0, #10
    bl    0 <rec_sum>
    str   r0, [r7, #4]
return 0;
movs  r3, #10
mov   r0, r3
adds  r7, #8
mov   sp, r7
pop   {r7, pc}
```

main() from (saved LR, Locals)
rec_sum(10), i=10, saved LR
rec_sum(9), i=9, saved LR
rec_sum(8), i=8, saved LR
:
rec_sum(2), i=2, saved LR
rec_sum(1), i=1, saved LR

Stack grows downwards

rec_sum(1) = 1
rec_sum(2) = 2+1 = 3
rec_sum(3) = 3+3 = 6
rec_sum(4) = 4+6 = 10
rec_sum(10) = 10+45 = 55

4)

```
int mysum(int a, int b)
{
    return a+b;
}

int main(void)
{
    int i=5, int j=10, sum=0;
    sum = mysum(i,j);
    return 0;
}
```

```
int (a, b)
push  {r7}
sub   SP, #12
add   r7, SP, #10
str   [r1, r7, #0]
      return a+b
ldr   r2 [r7, #a]
ldr   r3 [r7, #0]
add   r3, r2

mov   r0, r3
adds  r7, #12
mov   SP, #r7
pop   {r7}
bx    lr
```

```
void main()
{
push {r7, lr}
sub   SP, #16
add   r7, SP, #0
int i=5, j=10, sum=0
movs  r3, #5
str   r3 [r7, #12]
movs  r3  #10
str   r3 [r7, #8]
movs  r3 , #0
str   r3, [r7, #4]
sum = mySum(i,j)
ldr   r1  [r7, #8]
ldr   r0 [r7, #12]
bl    0 <mySum>
str   r0 [r7, #4]
return 0
```

```
mov   r0, r3
adds  r7, #16
mov   SP, r7
pop   {r7, PC}
```

sp+20	temp (0)
sp+16	i = 5
sp+12	j = 10
sp+8	sum = 0
sp+4	(padding / temp)
sp	(padding / temp)

Argument Passing

[sp+16] → i = 5 → LDR → R0 = 5

[sp+12] → j = 10 → LDR → R1 = 10

BL mysum

Compiler in mysum

LDR a → R0 = 5

LDR b → R1 = 10

↓

ADD R0, R1

↓

R0 = 15 (return value)

Back in main

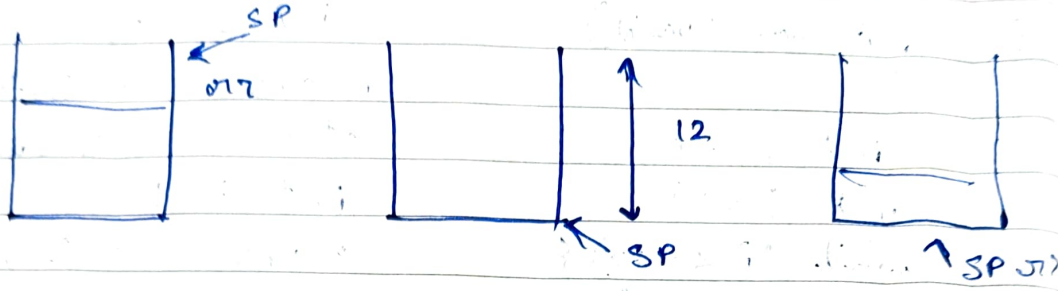
R0 = 15 → STR → [sp+8] sum = 15

5)

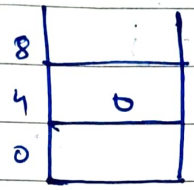
```
int main (void)
{
 int sum=0;
 int i=1;
while (i<=10){
      sum+=i;
       i++;
  }
return 0;
}
```

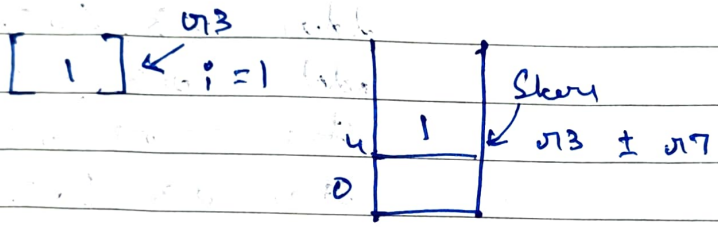
```
int sum=0;
movs    r3,#0;
str     r3,[r7,#4]
int i=1
movs    r3,#1
str     r3,[r7,#0]
while (i<=10)
b'n     le<main+0x1e>
ldr     r2,[r,#14]
ldr     r3,<r7,#0]
add     r3,r2
~~str     r3,r2~~
str     r3,[r7,#4]
```

```
i+=i
ldr  r2 [r7,#4]
ldr  r3 [r7,#0]
add  r3,r2
str  r3 [r7,#4]
 i++
ldr  r3 [r7,#0]
adds r3,#1
 str  r3 [r7,#10]
while (i<=10){
ldr  r3,[r7,#10]
cmp  r3,#10
ble.n 10<main+0x10>
} return 0;
movs r3 #0
mov  r0,r3
adds r7,#12
mov  SP,r7
pop  {r7}
bx   lr
```



r3 [0] = Sum





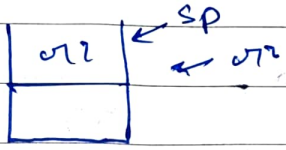
while loop ends

r2 = [1]
r3 = [0]
r3 [1]
r3 = r3 + 1

classmate
Date
Page

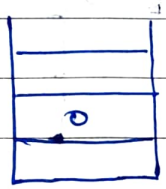
b) For loop

```
int main (void)
{
 int sum=0;
 int i ;
for (i=0; i<=10; i++){
    sum += i ;
 return 0;
}
```




```
push {r7}
 sub  sp #12;
```

```
add  r7, sp, #0
movs  r3 , #0
str   r3 [r7, #4]
```




```
for ( i =1 ; i<=10 ; i++)
movs   r3, #1          [1] -> r3
  bim   le <man oxle>
```

```
Sum = sum + 1
ldr   r3 [r7, #4]
ldr   r3 [r7, #0]
```






```
add   r3, r2      -> r3 = r3+r2
str   r3, r7, #4
```




```
for (i =1 ; i =10; i ++)
```

r3 [1]